F00-SR-001
PLANT 33kV HV DISTRIBUTION SWITCHROOM

FROM SWITCHROOM E-SHUT F00SR001-LCS (NOTE 8)
DRG 2000-F00-SLD-EL-10003

F00-SW-010
PLANT 33kV HV DISTRIBUTION SWITCHBOARD

CONT'D FROM SHEET 2 DRG 2000-F00-SLD-EL-10003

CONT'D ON SHEET 4 DRG 2000-F00-SLD-EL-10005

33kV INCOMER 2
FROM PARKESTON POWER STATION
FIM B (POLE FB42) BY KCGM

36kV 2000A 3-E 31.5kA FOR 3s

B12 | B13 | B14 | B15

630A | 2000A | 500/1A 2.5VA 5P20 | 300/1A 1VA 10P10 | 2000/1A 2.5VA CL0.2 | 2000/1A 2.5VA 5P20

CLOSE | TRIP | TEST | NOTE 2 | NOTE 1

PR | SEL 751 | SEL 787 | PQM | SEL 735

50 | 50G | 51 | 51G | 27 | 59 | AFD

INTERTRIP TO RECLOSER (REFER NOTE 9)

SWITCHROOM E-SHUT (REFER NOTE 8)

TRANSFORMER PROTECTION ALARM & TRIP CONTACTS (NOTE 6)

110V AC BUS VOLTAGE SUPPLY FROM INCOMER INC-2 BUS VT (PANEL B15)

VT
33kV / √3 / 0.11kV / √3 / 0.11kV / 3
35VA CLASS 0.2
35VA CLASS 3P

INC-2
2000A

F13TR401-JB
SAG MILL VSD TRANSFORMER JUNCTION BOX

TRANSFORMER PROTECTION DEVICES

3x2 3x2 3x2 3x2 | 3x2 3x2

F13-TR-401
33/3.8/3.8kV 33/3.8/3.8kV
33/0.415/0.415kV
Dyd Dyd Yyd - (NOTE 10)
TOTAL : 25400kVA
PRIMARY : 12000kVA
: 12000kVA
: 1400kVA
SECONDARY : 2x6000 kVA
: 2x6000 kVA
2x700 kVA
KNAN

CT's
1200/1A
5VA
SP20

TRANSFORMER SECONDARY CT's

SWITCHROOM E-SHUT (REFER NOTE 4)
CIRCUIT BREAKER TRIP COMMAND
CIRCUIT BREAKER CLOSE COMMAND
CIRCUIT BREAKER CLOSED SIGNAL
DISCONNECTOR STATUS SIGNAL
110V AC BUS VOLTAGE SUPPLY

ACTIVATION OF SWITCHROOM E-SHUT (REFER NOTE 4)
DETECTION OF ARC FAULT (REFER NOTE 5)

F13MS401A-VS
SAG MILL MOTOR 1 HV VSD (MASTER) 3.8/3.5kV

F13MS401B-VS
SAG MILL MOTOR 2 HV VSD (SLAVE) 3.8/3.5kV

SAG MILL MOTOR 1 EXCITER PANEL

SAG MILL MOTOR 2 EXCITER PANEL

SAG MILL VSD (MASTER) AUXILIARY PANEL

F13MS401AVS-ISL03 | F13MS401AVS-ISL01 | F13MS401AVS-ISL02 | F13MS401BVS-ISL01 | F13MS401BVS-ISL02 | F13MS401BVS-ISL03

ISOLATOR PANEL | ISOLATOR PANEL A | ISOLATOR PANEL B

F13-SR-003
SAG & SAG MILLS VSD SWITCHROOM

CONT'D FROM SHEET 2 DRG 2000-F00-SLD-EL-10003

EXCITER | M 3~ | 10000 kW | RTD

F13-MS-401A
SAG MILL MOTOR 1
2000-ME001.2-D02-50006

F13-MS-401B
SAG MILL MOTOR 2
2000-ME001.2-D02-50006

F13-TR-002
SAG MILL MCC 33/0.433kV TRANSFORMER

F13-SW-002
SAG MILL 415V MCC INCOMER

REFER DRG 2000-F13-SLD-EL-10301

33kV SPARE (2)
FULLY EQUIPPED
-

NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE SEL787 TRANSFORMER FEEDER PROTECTION RELAYS.
7. THE INTERTRIP OF THE 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
8. THE ACTIVATION OF THE PLANT 33kV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
9. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.
10. REFER TO NAMEPLATE DRAWING 2000-ME001.3-D05-50001 FOR LV WINDING PHASE SHIFT ANGLES.

LEGEND

- PR — INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
- PQM — POWER QUALITY AND REVENUE METERING
- ARC FLASH SENSOR (POINT TYPE) NOTE 2
- CAPACITIVE VOLTAGE INDICATION
- MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
- (27) UNDER-VOLTAGE
- (50) INSTANTANEOUS OVERCURRENT
- (50G) (50N) INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
- (51) TIMED OVERCURRENT
- (51G) (51N) TIMED EARTH/NEUTRAL OVERCURRENT
- (59) OVER VOLTAGE
- (64) EARTH FAULT
- (67) (67G) DIRECTIONAL OVERCURRENT/EARTH FAULT
- (AFD) ARC FLASH DETECTION

2000-F00-SLD-EL-10004_Rev5

Single line diagram sheet 3